BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 018 · Autumn 1 AP1 Assessment and Review

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does input validation check?

Q2. How can a larger cache help a CPU?
A. More physical CPU cores
B. Fewer slow memory accesses
C. All RAM becomes non-volatile
D. No instructions need fetching

Q3. Why is clock speed alone insufficient to compare two CPUs?

Q4. A quiz needs scores, questions, input and a menu. “Decomposition hides all parts in one unplanned step.” Correct this. How should the quiz be split up?

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding

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I Do · Read new knowledge

Key vocabulary

register — Small, fast CPU storage holding data or addresses.

compression — Reducing bits needed to represent a file.

abstraction — Ignoring irrelevant problem details.

validation — Checking data against specified input rules.

New learning

Paper 1 knowledge: Review binary, character sets, image/sound file sizes, compression, RAM/ROM, storage and CPU architecture.

Paper 2 introduction: Review abstraction, decomposition, IPO, flowcharts, searching, simple sorting, variables and validation.

Show working: For calculations, write formula, substitute values, calculate bits and convert units. For algorithms, show intermediate steps.

Use feedback: Identify one concept you can explain clearly and one that requires further teaching or practice.

Worked example

Model a synoptic answer: calculate bitmap storage and explain a suitable device for a portable laptop.

1. A 50 × 40 bitmap at 8-bit colour depth has 50 × 40 = 2,000 pixels.

2. Multiply by 8: 16,000 raw bits.

3. Divide by 8: 2,000 bytes of uncompressed pixel data.

4. For a laptop that may be knocked, select SSD for its lack of moving mechanical parts.

5. State one drawback, such as higher cost per GB, and explain why this is not universal.

We Do · Practise with guidance

Guided 1

A 5 × 4 bitmap at 8 bits per pixel uses how many bytes?

Worked method: 5 × 4 × 8 = 160 bits = 20 bytes.

Guided 2

Linear search for 6 in [2, 5, 6, 1]: give the zero-based index.

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Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Whether input meets specified rules; it does not prove factual truth or identity.

Q2. B — Fewer slow memory accesses

Q3. Core count, cache and workload also affect performance; comparable evidence is needed.

Q4. Split a large problem into smaller parts. Keep the parts needed for the task, then link clear steps to solve it.

I Do / We Do pages · Guided practice

Q1. 5 × 4 × 8 = 160 bits = 20 bytes.

Q2. Index 2.

Q3. PC holds next instruction address; repeated cycles process successive instructions.

You Do pages · Code First

Expected result: 10 × 5 × 8 = 400 bits. At 4-bit depth, 200 bits. Divide bits by 8: 25 bytes at 4-bit depth. Code is a small retrieval task, not a substitute for the assessment.

You Do pages · Independent questions

Q1. 100 × 20 = 2,000 pixels [1]; 2,000 × 4 = 8,000 bits [1]; 8,000 ÷ 8 = 1,000 bytes [1].

Q2. Abstraction ignores unnecessary detail [1], while keeping essential information [1]; storage capacity and file size matter, device colour does not [1].

Q3. Check mark >= 0 AND mark <= 100 [2]; boundary 0 or 100 [1], expected accepted [1].

Exit Ticket · Model answer

False

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Worked method: Index 2.

Guided 3

State what the PC holds and why the FDE cycle repeats.

Worked method: PC holds next instruction address; repeated cycles process successive instructions.

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the raw number of bits in a bitmap.

```
width = 10
height = 5
depth = 8
print(width * height * depth)
```

Predict the output.

Change depth to 4.

Explain how to convert the result to bytes.

Question 1

A 100 × 20 image uses 4 bits per pixel. Calculate its raw size in bytes. [3 marks]

Question 2

Explain abstraction and give an example of a detail that could be ignored in a storage-capacity calculator. [3 marks]

Question 3

A pupil enters a mark of 105 into a program accepting 0–100. Explain one validation rule and one suitable boundary test. [4 marks]

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Exit Ticket · Answer before checking

An unsigned 8-bit binary number can represent denary 300.